localCloudConfiguration

Service Description

Service ID: *"localCloudConfiguration"*

Abstract
This document describes the `localCloudConfiguration` service, version 1.0.0, produced by the LocalCloudConfigurator support system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope 3

2 Overview 3
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface 4
3.1 operation catalog . . . 4
3.2 operation status . . . 4
3.3 operation plan . . . 4
3.4 operation submit-deployment . . . 4
3.5 operation get-deployment . . . 4
3.6 operation list-configurations . . . 4
3.7 operation create-configuration . . . 4
3.8 operation get-configuration . . . 4
3.9 operation update-configuration . . . 4
3.10 operation remove-configuration . . . 5

4 Information Model 6
4.1 struct ApplyRequest . . . 6
4.2 struct Catalog . . . 6
4.3 struct Component . . . 6
4.4 struct Configuration . . . 6
4.5 struct ConfigurationDraft . . . 7
4.6 struct ConfigurationList . . . 7
4.7 struct Deployment . . . 7
4.8 struct DeploymentAccepted . . . 7
4.9 struct Plan . . . 7
4.10 struct PlanRequest . . . 8
4.11 struct PlanResponse . . . 8
4.12 struct Point . . . 8
4.13 struct Removal . . . 8
4.14 struct RevisionCheck . . . 8
4.15 struct Status . . . 8
4.16 struct Step . . . 8
4.17 struct StepState . . . 9
4.18 Primitives . . . 9

5 References 10

6 Revision History 11
6.1 Amendments . . . 11
6.2 Quality Assurance . . . 11

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `localCloudConfiguration` service, which lets a consumer read the component catalog and the runtime state of a local cloud, keep designs, review plans and apply them. It is produced by the LocalCloudConfigurator system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

The operations are those of the earlier operator API of the Java LocalCloudConfigurator (paths below `/api`), now offered as an Arrowhead service.

2.2 How This Service Is Meant to Be Used

LocalCloudManager, the consumer, shows the catalog and the live status in its graphical editor, saves designs, asks for a plan, shows it to the operator and applies it by its hash after confirmation; it then follows the deployment step by step.

2.3 Important Delimitations and Dependencies

Every operation needs an authenticated caller and Sysop, the configurator's whitelist or a ConsumerAuthorization grant of that operation. A plan is applied only while its hash matches a plan built afresh; one deployment runs at a time. Saving or deleting a design never changes the runtime. Completed steps of a failed deployment remain applied.

3 Service Interface

This section describes the operations of the `localCloudConfiguration` service. Each subsection names an abstract operation, an input type and an output type, in that order. All abstract data types named in this section are defined in Section 4.

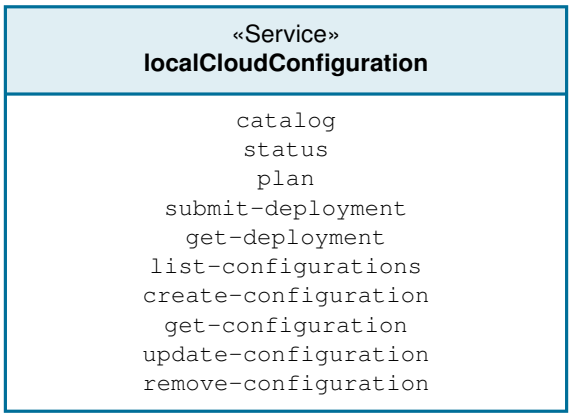


Figure 1: SysML-style block of the `localCloudConfiguration` service and its operations.

3.1 operation catalog () : Catalog

The component catalog of the cloud.

3.2 operation status () : Status

Running Compose services and the state (`running`, `partial`, `stopped`) of each component (cached 2 s).

3.3 operation plan (PlanRequest) : PlanResponse

Builds a plan against the current runtime; returns it with its hash.

3.4 operation submit-deployment (ApplyRequest) : DeploymentAccepted

Applies a reviewed plan if its hash still matches a fresh plan and no other deployment is running.

3.5 operation get-deployment (Identifier) : Deployment

State and step results of a submitted deployment (identifier appended to the path); the report is stored.

3.6 operation list-configurations () : ConfigurationList

Saved designs (up to 1000), sorted by name.

3.7 operation create-configuration (ConfigurationDraft) : Configuration

Saves a new design (revision 1).

3.8 operation get-configuration (Identifier) : Configuration

One saved design.

3.9 operation update-configuration (ConfigurationDraft) : Configuration

Replaces a design when `expectedRevision` matches; the revision is incremented.

3.10 operation remove-configuration (RevisionCheck) : Removal

Deletes a design when `expectedRevision` matches; the runtime is unchanged.

4 Information Model

Here, all data objects that can be part of the `localCloudConfiguration` service are listed in alphabetic order.

4.1 struct ApplyRequest

Field	Type	Description
systems	List<Name>	Selected components.
remove	List<Name>	Components to stop.
reapply	Boolean	Rerun setup of running components.
smoke	Boolean	Add the verification step.
expectedPlanHash	String	Hash of the reviewed plan (409 if it no longer matches).

4.2 struct Catalog

Field	Type	Description
cloud	Name	Local cloud.
composeProject	Name	Compose project.
colony	Name	ColonyOS colony (legacy backend).
executorType	Name	ColonyOS executor type.
defaults	List<Name>	Default core components.
smoke	List<Name>	Verification operations (`verify-localcloud`).
systems	List<Component>	Components.

4.3 struct Component

Field	Type	Description
id	Name	Component identifier.
name	String	Display name.
kind	String	`core`, `support`, `application` or `infrastructure`.
services	List<Name>	Compose services, databases first.
internalAssets	List<Name>	Owned databases.
requires	List<Name>	Dependencies.
provision	List<String>	Provisioning scripts.
policies	List<String>	Policy scripts.
protected	Boolean	Cannot be stopped.

4.4 struct Configuration

Field	Type	Description
id	Identifier	UUID.
name	String	1–200 characters.
systems	List<Name>	Defaults first, then the selection (databases mapped to their host).
remove	List<Name>	Components marked for stopping.
reapply	Boolean	
smoke	Boolean	
layout	Object<Point>	Positions by component.
revision	Number	Incremented on every update.
updatedAt	DateTime	Time of the last change.

4.5 struct ConfigurationDraft

Field	Type	Description
name	String	1–200 characters.
systems	List<Name>	
remove	List<Name>	
reapply	Boolean	
smoke	Boolean	
layout	Object<Point>	At most 200 positions.
expectedRevision	Number	Required for update-configuration.

4.6 struct ConfigurationList

Field	Type	Description
entries	List<Configuration>	Saved designs.

4.7 struct Deployment

Field	Type	Description
id	Identifier	Deployment identifier.
state	String	`running`, `success` or `failed` (ColonyOS: also `waiting`).
steps	List<StepState>	One entry per plan step.

4.8 struct DeploymentAccepted

Field	Type	Description
id	Identifier	Deployment identifier (empty when there was nothing to do).
plan	Plan	The applied plan.
state	String	`submitted`, or `success` when the plan had no steps.

4.9 struct Plan

Field	Type	Description
cloud	Name	Local cloud.
catalogHash	String	SHA-256 of the catalog.
systems	List<Name>	Components in dependency order.
steps	List<Step>	Ordered steps.
notes	List<String>	For example `"TimeManager: already running"`.

4.10 struct PlanRequest

Field	Type	Description
systems	List<Name>	Selected components (at most 200).
remove	List<Name>	Components to stop (at most 200).
reapply	Boolean	
smoke	Boolean	

4.11 struct PlanResponse

Field	Type	Description
plan	Plan	The plan.
hash	String	SHA-256 of the plan.

4.12 struct Point

Field	Type	Description
x	Number	0–10000.
y	Number	0–10000.

4.13 struct Removal

Field	Type	Description
removed	Identifier	Removed design.

4.14 struct RevisionCheck

Field	Type	Description
expectedRevision	Number	Current revision.

4.15 struct Status

Field	Type	Description
cloud	Name	Local cloud.
running	List<Name>	Running Compose services.
systems	List<Object>	`id` and `state` per component.

4.16 struct Step

Field	Type	Description
kind	String	`provision`, `up`, `policy`, `stop` or `smoke`.
target	String	Script, Compose service or verification.
system	Name	Owning component (absent for `smoke`).

4.17 struct StepState

Field	Type	Description
node	String	Step identifier (`s1`, ...).
processId	String	ColonyOS process (empty for the worker).
kind	String	Step kind.
target	String	Step target.
state	String	`waiting`, `running`, `success` or `failed`.
errors	List<String>	Redacted errors.
output	String	Redacted, bounded command output.

4.18 Primitives

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record.
List	An ordered collection of items of one type.
Name	A string identifier, human and machine readable.
Number	A number.
Object	A structured value.
String	A chain of characters.

5 References

[1] J. Delsing, “LocalCloudConfigurator – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-LocalCloudConfigurator-java/documentation/v5.3.0/LocalCloudConfigurator-v5.3.0-SysD.pdf.

[2] ——, “localCloudConfiguration HTTP/TLS/JSON – Interface Design Description (LocalCloudConfigurator),” Eclipse Arrowhead, IDD 5.3.0, 2026, localCloudConfigurator-v5.3.0-localCloudConfiguration-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-11-09	5.2.2	Document created for the `localCloudConfiguration` service of the Java-Spring LocalCloudConfigurator.	Claude Opus 5.5 (draft)
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	